PSEUDO-COHERENCE OF A PERFECT-TYPE COMPLEX ON AN AFFINE SCHEME IS A MODULE-LEVEL PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 2450 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Local characterization of m-pseudo-coherent objects. structural (definition unfolding). Standard opens with strictly perfect complexes approximating E in degrees $> m$.
- Refining maps to actual complex maps. Cohomology, lemma local-actual (cross-chapter, named). Turns the local quasi-isomorphisms into maps of complexes after refining the cover.
- Summands of finite locally free are finite locally free. Modules, lemma direct-summand-of-locally-free-is-locally-free (cross-chapter, named). The approximating terms are finite locally free, then finite free after refinement.
- Gluing pseudo-coherent complexes of modules. More on Algebra, lemma glue-pseudo-coherent (cross-chapter, named). Descends the module-level pseudo-coherence from the standard covers.
- m-pseudo-coherent iff m-pseudo-coherent for all m. More on Algebra, lemma pseudo-coherent (cross-chapter, named). Handles the pseudo-coherent (all m) case.

Source reference. Original reference 08C9: focused-seeds/geometry/S9-perf2-pseudo-coherent-affine/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BCI: focused-seeds/geometry/S9-perf2-pseudo-coherent-affine/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 066D: focused-seeds/geometry/S9-perf2-pseudo-coherent-affine/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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Source reference. Original reference 064U: focused-seeds/geometry/S9-perf2-pseudo-coherent-affine/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $X = \mathrm{Spec}(A)$ be an affine scheme. Let $M^\bullet$ be a complex of A-modules and let E be the corresponding object of $D(\mathcal{O}_X)$. Then E is an m-pseudo-coherent (resp. pseudo-coherent) as an object of $D(\mathcal{O}_X)$ if and only if $M^\bullet$ is m-pseudo-coherent (resp. pseudo-coherent) as a complex of A-modules.*

Proof. It is immediate from the definitions that if $M^\bullet$ is m-pseudo-coherent, so is E. To prove the converse, assume E is m-pseudo-coherent. As $X = \mathrm{Spec}(A)$ is quasi-compact with a basis for the topology given by standard opens, we can find a standard open covering $X = D(f_1) \cup \ldots \cup D(f_n)$ and strictly perfect complexes $\mathcal{E}_i^\bullet$ on $D(f_i)$ and maps $\alpha_i : \mathcal{E}_i^\bullet \to E|_{U_i}$ inducing isomorphisms on H^j for $j > m$ and surjections on H^m. By Cohomology, Lemma 08C9 after refining the open covering we may assume α_i is given by a map of complexes $\mathcal{E}_i^\bullet \to \widetilde{M^\bullet}|_{U_i}$ for each i. By Modules, Lemma 0BCI the terms $\mathcal{E}_i^n$ are finite locally free modules. Hence after refining the open covering we may assume each $\mathcal{E}_i^n$ is a finite free $\mathcal{O}_{U_i}$-module. From the definition it follows that $M_{f_i}^\bullet$ is an m-pseudo-coherent complex of A_{f_i}-modules. We conclude by applying More on Algebra, Lemma 066D.

The case "pseudo-coherent" follows from the fact that E is pseudo-coherent if and only if E is m-pseudo-coherent for all m (by definition) and the same is true for $M^\bullet$ by More on Algebra, Lemma 064U. □